Simple polygons whose sides lie on n lines: a parity obstruction and constructions

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Abstract

Let $M(n)$ be the largest number of sides of a simple polygon all of whose sides lie on n lines. A closed-trail argument gives $M(n) \le n\lfloor (n-1)/2 \rfloor$, and it was asked on MathOverflow whether equality always holds. For odd $n = 2r+1$ we prove two things. First, in every simple arrangement of n pseudolines, a certain 2-regular graph on the intersection points has a number of components congruent to $(n^2-1)/8 \pmod 2$. Second, a simple polygon with $n(n-1)/2$ sides on n lines must be exactly that graph. Consequently, if $n \equiv \pm 1 \pmod 8$ then $M(n) \le n(n-1)/2 - 3$; in particular the conjectured equality fails for $n = 7, 9, 15, 17, \ldots$. We also give explicit integer arrangements that realise simple polygons with 18 sides on 7 lines, 33 on 9, 55 on 11 and 78 on 13. Hence $M(7) = 18$ and $M(9) = 33$, while $M(11) = 55$ and $M(13) = 78$ attain the bound. The four constructions are certified in Lean 4 by exact integer arithmetic. The parity theorem and the upper bounds are proved on paper only; they are confirmed by exhaustive computation for $n \le 9$ but are not formally verified.

Contents

1 Introduction **1**

2 Definitions and the reduction **3**

3 The parity theorem **4**

4 Constructions **6**

5 Formal verification of the constructions **7**

6 What is not verified; open questions **10**

A Vertex cycles of the larger constructions **11**

1 Introduction

A polygon is *simple* if its boundary is a Jordan curve. Suppose a simple polygon has k sides $S_1, \ldots, S_k$ and each side S_j lies on a line L_j taken from a fixed set of n lines. How large can k be? We write $M(n)$ for the maximum.

This question was asked on MathOverflow by Johnathan Will (https://mathoverflow.net/users/596423) on July 31, 2026 [3]. Their upper bound comes from the following observation. Consecutive sides lie on distinct lines, and since the vertex $L_j \cap L_{j+1}$ is visited only once, no unordered pair of lines occurs twice as a consecutive pair. So the cyclic sequence $L_1, \dots, L_k$ is a closed trail in the complete graph K_n, and k is at most the maximum length of a circuit in K_n, which is $n\lfloor (n-1)/2 \rfloor$ (OEIS A050187 [5]).

The asker conjectured that $M(n) = n\lfloor (n-1)/2 \rfloor$ for every n. They gave examples for $n \le 6$, using the pentagram outline for $n = 5$ and the Star of David for $n = 6$. In the comments, the user DSM (https://mathoverflow.net/users/12613) found a 24-gon on 8 lines and remarked that they could reach 18 sides with 7 lines but not 21. The asker reported that their own construction fails for $n = 11$, and asked whether one can prove that for $n = 7$ only a self-intersecting ("genus 1") polygon can reach 21 sides. As of October 8, 2026 the question had no answers. To our knowledge, as of that date, the results below do not appear in the literature.

Results. Write $n = 2r + 1$ for odd n; then $n\lfloor (n-1)/2 \rfloor = \binom{n}{2}$.

Theorem 1.1 (Parity obstruction)**.** *Let $n = 2r + 1$. In every simple arrangement of n lines, and more generally of n pseudolines, the* alternating 2-factor *G (Definition 2.2) has a number of connected components congruent to $r(r+1)/2 = (n^2-1)/8$ modulo 2.*

Theorem 1.2 (Reduction)**.** *If n is odd and a simple polygon with $\binom{n}{2}$ sides has its sides on n lines, then the arrangement is simple and the polygon is exactly the alternating 2-factor, which must therefore be connected. If instead such a polygon has $k \ne \binom{n}{2}$ sides, then $k \le \binom{n}{2} - 3$.*

Corollary 1.3. *If $n \equiv \pm 1 \pmod 8$, then $M(n) \le \binom{n}{2} - 3$. In particular $M(7) \le 18$ and $M(9) \le 33$, so the conjectured equality fails for every $n \equiv \pm 1 \pmod 8$.*

Theorem 1.4 (Constructions; Lean-certified)**.** *There are explicit arrangements of 7, 9, 11 and 13 lines with integer coefficients carrying simple polygons with 18, 33, 55 and 78 sides respectively (Section 4). Consequently*

$$M(7) = 18, \qquad M(9) = 33, \qquad M(11) = 55 = 11 \cdot 5, \qquad M(13) = 78 = 13 \cdot 6.$$

The values $M(11)$ and $M(13)$ follow from the constructions together with the elementary upper bound. The values $M(7)$ and $M(9)$ also need Corollary 1.3. The value $M(7) = 18$ confirms DSM's experimental 18 and shows that it is optimal.

Verification status.

- The four constructions of Theorem 1.4 are checked in Lean 4.34.1 (core only, no Mathlib) by kernel evaluation with exact integer arithmetic. The relevant theorems are `mo513798_n7`, `_n9`, `_n11` and `_n13`; the only axiom used is `propext`. See Section 5.
- Theorems 1.1 and 1.2 and Corollary 1.3 are proved on paper and *not* formally verified. The proof of Theorem 1.1 uses the Matsumoto–Tits theorem on reduced words. For $n \le 9$ the conclusions were also confirmed by exhaustive computation over all simple pseudoline arrangements; this is evidence, not a proof (Section 6).

Organisation. Section 2 proves Theorem 1.2. Section 3 proves Theorem 1.1. Section 4 gives the constructions and Section 5 their formal certification. Section 6 lists what is not verified and some open questions.

2 Definitions and the reduction

Definition 2.1. A *simple polygon with k sides* ($k \geq 3$) is a cyclic sequence of points $P_0, \ldots, P_{k-1}$ in the plane with two properties. First, P_{j-1}, P_j, P_{j+1} are never collinear (indices mod k), so there are genuine corners. Second, any two non-adjacent sides $[P_i, P_{i+1}]$ and $[P_j, P_{j+1}]$ are disjoint closed segments. Its sides *lie on* a set $\mathcal{L}$ of lines if every side is contained in some line of $\mathcal{L}$. Let $M(n)$ be the largest k for which a simple polygon with k sides has its sides on some set of n lines.

An arrangement of lines is *simple* if no two lines are parallel and no three are concurrent.

Definition 2.2 (Alternating 2-factor)**.** Let $\mathcal{A}$ be a simple arrangement of an odd number n of lines. On each line ℓ, list the $n-1$ intersection points in their order along ℓ as $p_1, \ldots, p_{n-1}$. The graph $G(\mathcal{A})$ has the $\binom{n}{2}$ intersection points as vertices, and its edges are the pairs

$$\{p_1, p_2\},\ \{p_3, p_4\},\ \ldots,\ \{p_{n-2}, p_{n-1}\}$$

on every line.

Every vertex lies on exactly two lines and gets one edge from each of them, so G is 2-regular, i.e. a disjoint union of cycles.

Proposition 2.3. *Let n be odd and let a simple polygon with k sides have its sides on n lines.*

1. *If $k = \binom{n}{2}$, then the arrangement is simple and the polygon's sides are exactly the edges of G (drawn as segments). In particular G is a single cycle.*
2. *If $k \neq \binom{n}{2}$, then $k \leq \binom{n}{2} - 3$.*

Proof. Let the sides be $S_j \subset L_j$ ($j \in \mathbb{Z}/k$), with $L_j \neq L_{j+1}$ and vertices $V_j = L_j \cap L_{j+1}$. The vertices are distinct, so the pairs $\{L_j, L_{j+1}\}$ are pairwise distinct. These pairs form a set E' of k edges of K_n, and the cyclic sequence (L_j) is a closed trail through them. Hence every vertex of K_n has even degree in E'.

Part 2. K_n is $(n-1)$-regular and $n-1$ is even, so the complement $E(K_n) \setminus E'$ also has all degrees even. If the complement is non-empty it contains a cycle, so it has at least 3 edges.

Part 1. Now $E' = E(K_n)$. Every two lines meet, so no two are parallel. The $\binom{n}{2}$ points V_j are distinct, so no three lines are concurrent. Thus the arrangement is simple, and every intersection point is a vertex.

Fix a line ℓ with intersection points $p_1, \ldots, p_{n-1}$ in order. Each $p_i = \ell \cap m$ is a vertex where a side on ℓ meets a side on m. So each p_i is an endpoint of exactly one side contained in ℓ, and there are $(n-1)/2$ such sides. Two sides on ℓ are never consecutive, since consecutive sides lie on different lines. Hence they are pairwise disjoint closed segments.

The side with endpoint p_1 ends at some p_m. If $m > 2$, its relative interior contains p_2, which is an endpoint of another side on ℓ; this contradicts disjointness. So $[p_1, p_2]$ is a side. Repeating the argument on $p_3, \ldots$ shows that the sides on ℓ are exactly $[p_1, p_2], [p_3, p_4], \ldots$, which are the edges of G on ℓ. The polygon visits all $\binom{n}{2}$ vertices of G along edges of G, so G is one cycle. □

Remark 2.4. Conversely, if G is a single cycle then it is the boundary of a simple polygon with $\binom{n}{2}$ sides. We do not need this, because our constructions are verified directly.

Wiring diagrams. Rotate the plane so that no line is vertical and all intersection points have distinct x-coordinates. Number the lines $1, \dots, n$ from top to bottom far to the left. A vertical line sweeping from left to right sees the vertical order of the lines change exactly at intersection points, each time by swapping the lines at two adjacent heights $i, i+1$. This gives a word $i_1 i_2 \cdots i_N$ with $N = \binom{n}{2}$ in which every pair of lines (*wires*) swaps exactly once, i.e. a reduced word for the longest permutation w_0 of S_n [2].

The order of the intersection points along a line equals the order of the corresponding letters in the word. So G depends only on the word, and we define G in the same way for any reduced word of w_0, that is, for any simple arrangement of pseudolines. All arguments in Section 3 use only the word.

3 The parity theorem

Fix $n = 2r + 1$ and a reduced word $w = i_1 \cdots i_N$ of $w_0 \in S_n$. Wire x starts at height x (heights $1, \dots, n$ from top to bottom). Letter i_t is the *crossing* t: the wires at heights i_t and $i_t + 1$ exchange heights. Along each wire, its $n - 1$ crossings are numbered $1, \dots, n - 1$ in time order. These are its *positions*.

Definition 3.1. A crossing is an *A-crossing* if it has an even position along both of its wires. Equivalently, both wires have made an odd number of crossings before it. Let $\#A(w)$ be the number of A-crossings.

In G every edge joins an odd position to the next even position on one wire. So an A-crossing is a vertex both of whose G-neighbours occur earlier in time.

Lemma 3.2. *Every cycle of G contains an odd number of A-crossings. Consequently the number of components of G is congruent to $\#A(w)$ modulo 2.*

Proof. Process the crossings in time order. After t steps, call a wire *open* if it has made an odd number of crossings. Its last crossing is then waiting for its G-partner along that wire.

Let G_t be the subgraph of G formed by the processed crossings and the edges with both ends processed. Every component of G_t is a cycle of G or a path. Each path has two *dangling ends*, and each dangling end belongs to an open wire; distinct open wires carry distinct dangling ends. A path consisting of a single vertex has both dangling ends at that vertex.

Pairing the two dangling ends of each path gives a perfect matching μ_t on the set S_t of heights currently occupied by open wires. Consider crossing $t + 1$, between wire u at height i and wire v at height $i + 1$, and call it τ. Exactly one of the following happens.

(B) *Both wires closed.* τ has an odd position on both wires. It starts a new path whose dangling ends now sit at heights $i, i + 1$, so the pair $(i, i + 1)$ is added to μ.

(C) *Exactly one wire open.* Say u is open; the case of v is symmetric. Then τ extends the path that dangled on u. Its new dangling end is on v, which now occupies height i, the height u had. So S and μ are unchanged.

(A) *Both wires open.* τ is an A-crossing and joins the dangling ends at heights i and $i + 1$. If $\mu(i) = i + 1$ this *closes* a cycle of G. Otherwise it *joins* two paths: remove $(i, \mu(i))$ and $(i + 1, \mu(i + 1))$ and add $(\mu(i), \mu(i + 1))$.

Wires not involved in τ do not move.

By induction on t we prove two invariants.

(I1) μ_t is non-crossing for the order of heights. Case (B) adds a pair of adjacent heights, which cannot interleave with any pair. In case (A) the pairs $(i, \mu(i))$ and $(i+1, \mu(i+1))$ do not cross and $i, i+1$ are adjacent. So they lie either side by side, $\mu(i) < i < i+1 < \mu(i+1)$, or nested, $i < i+1 < \mu(i+1) < \mu(i)$ or its mirror image. A pair crossing the merged pair would cross one of the two old pairs.

(I2) For every pair $(a, b) \in \mu_t$ with $a < b$, the difference $b - a$ is odd. This is true for $(i, i+1)$. Under a side-by-side join, $b - a = (i - a) + 1 + (b - i - 1)$ is a sum of two odd numbers plus 1. Under a nested join, the new difference is (odd)−(odd)−1.

Fix a cycle C of G and let $\Lambda_C(t) = \sum(b - a - 1)/2$, the sum over the pairs of μ_t whose path lies in C. By (I2) this is an integer. We compare Λ_C before and after each step that involves C:

- a (B)-step adds 0;
- a (C)-step changes nothing;
- closing removes $(i, i+1)$, so it changes Λ_C by 0;
- a side-by-side join changes it by $\big((b - a - 1) - (i - a - 1) - (b - i - 2)\big)/2 = 1$;
- a nested join $(i, a) \supset (i+1, b)$ changes it by $-(b - i - 1)$, which is odd by (I2);
- the mirror nested join $(b, i+1) \supset (a, i)$ changes it by $a - i$, again odd.

Steps at crossings not on C do not touch the pairs of C.

The earliest vertex of C has both neighbours later in time, so it is a (B)-vertex and $\Lambda_C = 0$ right after it. Just before C closes, the only pair of C is $(i, i+1)$, so again $\Lambda_C = 0$. Each closing step closes exactly one cycle, so C has exactly one closing step. Therefore C has an even number of joins, and its A-crossings (joins plus one closing) are odd in number. Summing over the cycles gives the congruence. □

Lemma 3.3. *For every reduced word w of $w_0 \in S_{2r+1}$ we have $\#A(w) \equiv r(r+1)/2 \pmod 2$.*

Proof. A wire changes height by ± 1 at each of its crossings. So the number of crossings it has made so far is congruent modulo 2 to its current height minus its label.

Consider a crossing at heights $(i, i+1)$ of wires u and v. They have not crossed before, so their initial order persists and $u < v$. The crossing is an A-crossing iff $i - u$ and $i + 1 - v$ are both odd. If $u + v$ is even this is impossible. If $u + v$ is odd it holds iff $i - u$ is odd.

Commutation moves. Exchanging adjacent letters i_t, i_{t+1} with $|i_t - i_{t+1}| \geq 2$ changes neither the wires nor the heights of any crossing.

Braid moves. Replacing $i, i+1, i$ by $i+1, i, i+1$ involves three wires $x < y < z$. Each of the three crossings xy, xz, yz keeps its wires, and its upper height changes by exactly 1. So the move flips the A-status of exactly those of the three pairs whose labels have odd sum. Among three integers the number of such pairs is 0 or 2, so $\#A$ changes by an even number.

Connectivity. By the theorem of Matsumoto and Tits [4], any two reduced words of w_0 are connected by commutation and braid moves. Hence $\#A \bmod 2$ is the same for all reduced words of w_0.

Computation. Take the bubble-sort word $(s_1 s_2 \cdots s_{n-1})(s_1 \cdots s_{n-2}) \cdots (s_1)$. Here every wire meets the other wires in increasing order of labels. For $u < v$ the crossing therefore has position $v - 1$ on u and position u on v. It is an A-crossing iff u is even and v is odd. Hence

$$\#A = \sum_{j=1}^{r} \#\{v \text{ odd} : 2j < v \leq 2r+1\} = \sum_{j=1}^{r} (r - j + 1) = \frac{r(r+1)}{2}.$$ □

Proof of Theorem 1.1. Combine Lemmas 3.2 and 3.3, and note that $(n^2-1)/8 = r(r+1)/2$. □

Proof of Corollary 1.3. For $n = 2r+1$, the number $r(r+1)/2$ is even iff $r \equiv 0, 3 \pmod 4$, i.e. iff $n \equiv \pm 1 \pmod 8$. In that case G has an even number of components, so it is not a single cycle. By Proposition 2.3(1), no simple $\binom{n}{2}$-gon lies on n lines, and by Proposition 2.3(2) every such polygon has at most $\binom{n}{2} - 3$ sides. □

Remark 3.4. For $n \equiv \pm 3 \pmod 8$, Theorem 1.1 only says that G has an odd number of components. It does not guarantee a single cycle; for example, both 1 and 3 components occur for $n = 5$.

4 Constructions

We write the line ℓ_m as $a_m x + b_m y = c_m$ with integer coefficients. Each polygon is given by the cyclic sequence of its vertices, where the vertex "p/q" is $\ell_p \cap \ell_q$. Consecutive vertices share a line, which carries the side between them. All data, including vertex coordinates as exact rationals, are in `data/polyN.json` and `lean/Main.lean`.

n	lines (a_m, b_m, c_m), $m = 1, \dots, n$
7	$(-6,17,-74)$, $(0,-19,-95)$, $(-19,14,-98)$, $(4,-7,8)$, $(-19,13,-44)$, $(8,11,41)$, $(-6,2,-41)$
9	$(-22,6,-118)$, $(-14,-23,103)$, $(18,-2,91)$, $(11,-6,-43)$, $(-24,1,-136)$, $(27,23,49)$, $(-3,8,-149)$, $(14,-2,-14)$, $(16,21,-33)$
11	$(-39,31,-51)$, $(1,50,7)$, $(-22,45,-264)$, $(12,-49,-212)$, $(-10,-49,-25)$, $(-16,48,31)$, $(48,-14,68)$, $(-19,46,-80)$, $(-50,3,89)$, $(17,47,93)$, $(15,48,37)$
13	$(5,-19,255)$, $(17,11,229)$, $(-4,20,-1118)$, $(14,-14,106)$, $(19,4,126)$, $(3,-20,-505)$, $(-6,19,-402)$, $(10,17,78)$, $(20,-3,-68)$, $(-1,20,240)$, $(18,-8,29)$, $(7,-19,-397)$, $(-20,-3,-59)$

Table 1: The certified arrangements.

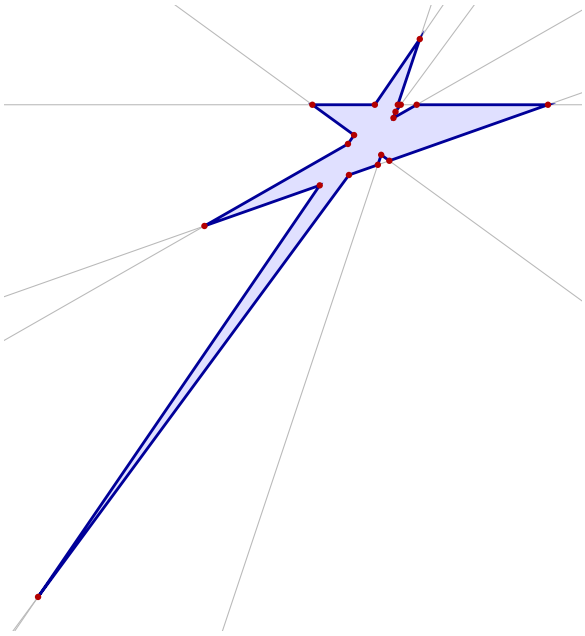
$n = 7$, an 18-gon. The vertex cycle is

$$1/2,\ 1/6,\ 6/7,\ 1/7,\ 1/3,\ 3/5,\ 1/5,\ 1/4,\ 4/5,\ 5/6,\ 2/6,\ 2/5,\ 5/7,\ 2/7,\ 2/3,\ 3/7,\ 4/7,\ 2/4.$$

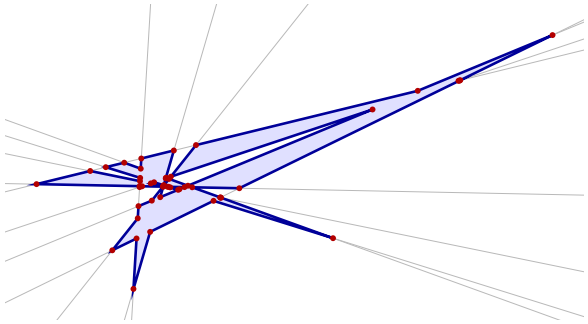
It uses every intersection point except the three points of the triangle of lines $\{3,4,6\}$. Lines $3,4,6$ each carry 2 sides and the other lines carry 3. The line pairs used form K_7 minus a triangle, which is the extremal case of Proposition 2.3(2).

$n = 9$, a 33-gon. This uses all intersection points except those of the triangle of lines $\{1,2,9\}$; the cycle is listed in Section A.

$n = 11$ and $n = 13$. Here the polygon is the alternating 2-factor G itself, which is a single cycle of length 55, respectively 78; it is listed in Section A. Each line carries $(n-1)/2$ sides.



18 sides on 7 lines



55 sides on 11 lines

Figure 1: Two of the certified polygons, drawn to scale. The grey lines are the arrangement, clipped to a box around the polygon. Several sides are very short, so some corners are not resolved at this scale; exact coordinates are in the data files.

How they were found. For $n = 11, 13$ we ran a simulated-annealing search over random real arrangements, minimising the number of components of G. Theorem 1.1 predicts an odd number for these n, and the search eventually reaches one component. The resulting coefficients were rounded to small integers and re-checked with exact rational arithmetic.

For $n = 7, 9$ we sampled random integer arrangements and, for each triangle $\{a, b, c\}$ of lines, formed the graph that pairs consecutive remaining intersection points on every line, leaving out $\ell_a \cap \ell_b$, $\ell_b \cap \ell_c$ and $\ell_a \cap \ell_c$. We kept the arrangement when this graph was a single cycle bounding a simple polygon.

The search method plays no role in correctness. Every polygon is verified directly, as described in Section 5.

Proof of Theorem 1.4. The four polygons are simple and have their sides on the listed lines; this is certified by the Lean theorems of Section 5. The lower bounds $M(n) \geq 18, 33, 55, 78$ follow.

For the matching upper bounds: $M(11) \leq 55$ and $M(13) \leq 78$ hold by the closed-trail bound $n\lfloor (n-1)/2 \rfloor$. $M(7) \leq 18$ and $M(9) \leq 33$ hold by Corollary 1.3. □

5 Formal verification of the constructions

The file `lean/Main.lean` is self-contained and was checked with Lean 4.34.1 using only the core library and Std; Mathlib is not used. The check took 37–53 s of wall time and about 2 GB of memory. The main statement is the following.

```
theorem mo513798_n11 :
    ∃ (L : List Ln) (P : List Pt),
      L.length = 11 ∧ P.length = 55 ∧ genuineLines L = true ∧
```

```
    isSimplePolygon P = true ∧ sidesOnLines L P = true :=
  ⟨lines11, poly11, by decide +kernel, by decide +kernel, by decide +kernel,
    by decide +kernel, by decide +kernel⟩
```

The theorems `mo513798_n13`, `mo513798_n7` and `mo513798_n9` have the same form, with $(n, k) = (13, 78)$, $(7, 18)$ and $(9, 33)$. All Boolean checks are discharged by `decide +kernel`, i.e. by evaluation in the kernel. No `sorry`, `axiom`, `admit` or `native_decide` occurs. The verbatim output of `#print axioms` (`lean/lean_output.txt`) is:

```
'mo513798_n11' depends on axioms: [propext]
'mo513798_n13' depends on axioms: [propext]
'mo513798_n7' depends on axioms: [propext]
'mo513798_n9' depends on axioms: [propext]
```

`propext` is a standard axiom of Lean. It enters through core decidability instances.

Definitions used by the statement (verbatim, doc-comments removed):

```
structure Pt where
  x : Int
  y : Int
  w : Int
deriving Repr

structure Ln where
  a : Int
  b : Int
  c : Int
deriving Repr

def onLine (l : Ln) (p : Pt) : Bool := l.a * p.x + l.b * p.y == l.c * p.w

def orient (p q r : Pt) : Int :=
  p.x * (q.y * r.w - q.w * r.y) - p.y * (q.x * r.w - q.w * r.x) + p.w * (q.x * r.y - q.y * r.x)

def fle (u uw v vw : Int) : Bool := decide (u * vw ≤ v * uw)

def inBox (p q r : Pt) : Bool :=
  ((fle p.x p.w r.x r.w && fle r.x r.w q.x q.w) || (fle q.x q.w r.x r.w && fle r.x r.w p.x p.w))
    &&
  ((fle p.y p.w r.y r.w && fle r.y r.w q.y q.w) || (fle q.y q.w r.y r.w && fle r.y r.w p.y p.w))

def segInter (a b c d : Pt) : Bool :=
  let d1 := orient c d a
  let d2 := orient c d b
  let d3 := orient a b c
  let d4 := orient a b d
  ((decide (d1 > 0) && decide (d2 < 0)) || (decide (d1 < 0) && decide (d2 > 0))) &&
    ((decide (d3 > 0) && decide (d4 < 0)) || (decide (d3 < 0) && decide (d4 > 0))) ||
  (d1 == 0 && inBox c d a) || (d2 == 0 && inBox c d b) ||
  (d3 == 0 && inBox a b c) || (d4 == 0 && inBox a b d)

def vtx (P : List Pt) (j : Nat) : Pt := P.getD (j % P.length) ⟨0, 0, 1⟩

def isSimplePolygon (P : List Pt) : Bool :=
  let k := P.length
  decide (3 ≤ k) &&
```

```
  P.all (fun p => decide (p.w > 0)) &&
  (List.range k).all (fun j => orient (vtx P j) (vtx P (j + 1)) (vtx P (j + 2)) != 0) &&
  (List.range k).all (fun i => (List.range k).all (fun j =>
    if decide (i + 2 ≤ j) && decide (j + 2 ≤ k + i) then
      !(segInter (vtx P i) (vtx P (i + 1)) (vtx P j) (vtx P (j + 1)))
    else true))

def sidesOnLines (L : List Ln) (P : List Pt) : Bool :=
  (List.range P.length).all (fun j =>
    L.any (fun l => onLine l (vtx P j) && onLine l (vtx P (j + 1))))

def genuineLines (L : List Ln) : Bool := L.all (fun l => l.a != 0 || l.b != 0)
```

Statement audit. We explain why the Lean statement means exactly "there are n lines and a simple polygon with k sides, every side lying on one of the lines".

- *Points.* A point is an integer triple (x, y, w); it represents $(x/w, y/w) \in \mathbb{Q}^2$. `isSimplePolygon` checks $w > 0$ for every vertex.
- *Lines.* `onLine` tests $ax + by = cw$, which is $a\frac{x}{w} + b\frac{y}{w} = c$ multiplied by $w > 0$. `genuineLines` excludes the degenerate equation $0x + 0y = c$, which would contain every point when $c = 0$.
- *Orientation.* `orient` is the 3×3 determinant of the homogeneous coordinates. It equals $p_w q_w r_w \cdot ((q - p) \times (r - p))$ for the affine points. Since the weights are positive, its sign is the orientation of (p, q, r), and it vanishes iff the points are collinear.
- *Segment intersection.* `segInter` is the standard exact test for closed segments [1, §33.1]: a proper crossing, or an endpoint of one segment lying on the other. The box test `inBox` is used only when an orientation is 0, i.e. for collinear points. In that case "lies in the bounding box" is equivalent to "lies on the segment". `fle` compares fractions with positive denominators.
- *Simple polygon.* Vertex j is `P[j mod k]`. `isSimplePolygon` requires:
  - $k \geq 3$;
  - every triple of consecutive vertices is non-collinear;
  - for $i < j$, the sides i and j are disjoint whenever $i + 2 \leq j$ and $j + 2 \leq k + i$.

  The last condition ranges over exactly the non-adjacent pairs of sides; the adjacent pairs are $j = i + 1$ and $(i, j) = (0, k - 1)$. Non-collinearity of consecutive triples makes consecutive vertices distinct and makes adjacent sides meet only in their common vertex. It also forbids two collinear consecutive sides, so the polygon has exactly k genuine sides. These conditions say exactly that the boundary is a Jordan polygon with k sides, as in the definition in Section 2.
- *Sides on lines.* `sidesOnLines` asks that both endpoints of each side lie on a common line of the list. Then the whole side lies on that line. The two length conditions fix n and k. If two listed lines coincided, the statement would only become stronger. In fact the listed lines are pairwise distinct and in general position; this was checked separately in exact rational arithmetic.
- *Checker sanity.* `Main.lean` also contains `example`s showing that the checker behaves as intended:

– it accepts a unit square;

– it rejects a bow-tie, three collinear consecutive vertices, and a polygon that touches itself;

– collinear segments are treated correctly whether they overlap or are disjoint.

Reproduction. Run `attempts/bin/leanchk lean/Main.lean` (the guarded wrapper around the Lean binary). Independently of Lean, the Python scripts in the accompanying directory `A09-polygon-lines/` recheck simplicity with exact rationals.

6 What is not verified; open questions

Not formally verified. The following are proved on paper only.

- Proposition 2.3, including the reduction from line arrangements to wiring diagrams.
- Lemma 3.2, the sweep argument.
- Lemma 3.3, which relies on the Matsumoto–Tits theorem [4].
- Hence Theorem 1.1, Corollary 1.3, and the upper bounds $M(7) \le 18$ and $M(9) \le 33$.

An attempt to certify Lemma 3.3 for $n = 7$ in Lean, by a kernel-evaluated breadth-first search over the sweep states of all 5040 permutations, ran out of kernel memory. We chose not to fall back on `native_decide`.

Computational evidence. This is not a proof. The sweep process in the proof of Lemma 3.2 defines a finite-state dynamic program. We validated it against direct component counts of G on random reduced words for $n = 3, 5, \dots, 11$, and then ran it over *all* reduced words of w_0 for $n \le 9$. For $n = 7$ and $n = 9$ the graph G is never a single cycle; it has 2 or at least 3 components. In every case $\#A$ had the same parity for all words, as Lemma 3.3 predicts.

On 2000 random reduced words for each odd $n \le 19$, the parity of the number of components always equalled $(n^2 - 1)/8 \bmod 2$. On 2500 random words (500 for each $n = 5, 7, 9, 11, 13$), every cycle contained an odd number of A-crossings. Random real arrangements of 7 lines produced 2, 4 or 6 components, never an odd number.

Open questions.

1. Is $M(n) = \binom{n}{2}$ for every $n \equiv \pm 3 \pmod 8$? We confirmed this for $n = 3, 5, 11, 13$. Brief searches for $n = 19$ and $n = 21$ (four runs of 10^6 annealing steps each) reached 3 components but not 1.
2. Is $M(n) = \binom{n}{2} - 3$ for every $n \equiv \pm 1 \pmod 8$? We confirmed this for $n = 7$ and $n = 9$.
3. What is $M(n)$ for even n? The closed-trail bound is $n(n-2)/2$, and DSM's 24-gon on 8 lines attains it for $n = 8$. We have not studied even n.
4. Is there a proof of Lemma 3.3 that avoids the Matsumoto–Tits theorem, for example via an explicit potential function on permutations? A linear-algebra search over GF(2) suggests that such potentials exist.

References

[1] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, 3rd edition, 2009.

[2] Jacob E. Goodman and Richard Pollack. Semispaces of configurations, cell complexes of arrangements. *J. Combin. Theory Ser. A*, 37(3):257–293, 1984.

[3] Johnathan Will. What is the maximum k for a k-gon inducible with an n-line arrangement? MathOverflow question 513798, https://mathoverflow.net/questions/513798 (posted July 31, 2026; comments by DSM, https://mathoverflow.net/users/12613), 2026. Accessed October 8, 2026.

[4] Hideya Matsumoto. Générateurs et relations des groupes de Weyl généralisés. *C. R. Acad. Sci. Paris*, 258:3419–3422, 1964.

[5] OEIS Foundation Inc. Sequence A050187: $a(n) = n\lfloor(n-1)/2\rfloor$ (maximum length of a circuit in the complete graph on n vertices). The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A050187. Accessed October 8, 2026.

A Vertex cycles of the larger constructions

The notation "p/q" means $\ell_p \cap \ell_q$, with lines numbered as in Table 1. Each list is the cyclic order of the vertices.

$n = 9$ (33 vertices; lines $1, 2, 9$ form the omitted triangle). 1/3, 1/6, 3/6, 3/9, 5/9, 5/6, 6/9, 7/9, 2/7, 2/6, 6/7, 5/7, 2/5, 2/3, 3/7, 1/7, 1/8, 7/8, 4/7, 2/4, 2/8, 8/9, 4/9, 4/6, 6/8, 4/8, 4/5, 5/8, 3/8, 3/4, 1/4, 1/5, 3/5.

$n = 11$ (55 vertices). 1/2, 1/8, 8/9, 1/9, 1/3, 3/9, 7/9, 3/7, 3/11, 10/11, 5/10, 3/5, 3/10, 2/10, 2/3, 3/4, 4/6, 3/6, 3/8, 4/8, 1/4, 1/6, 6/8, 8/10, 1/10, 1/11, 2/11, 2/8, 5/8, 5/11, 8/11, 7/8, 2/7, 2/5, 5/7, 1/7, 1/5, 5/6, 6/11, 7/11, 6/7, 6/10, 7/10, 4/7, 4/9, 9/10, 4/10, 4/11, 9/11, 5/9, 4/5, 2/4, 2/9, 6/9, 2/6.

$n = 13$ (78 vertices). 1/2, 1/5, 4/5, 4/8, 5/8, 5/11, 8/11, 8/13, 11/13, 4/11, 4/13, 1/13, 1/11, 9/11, 1/9, 1/4, 4/9, 8/9, 8/10, 9/10, 9/12, 8/12, 6/8, 6/10, 10/12, 1/12, 1/3, 3/12, 7/12, 3/7, 3/4, 4/7, 7/11, 3/11, 3/9, 7/9, 7/13, 3/13, 3/5, 5/7, 2/7, 2/3, 3/8, 7/8, 7/10, 3/10, 3/6, 1/6, 1/7, 6/7, 4/6, 4/12, 6/12, 6/11, 11/12, 5/12, 5/6, 6/9, 5/9, 5/13, 6/13, 2/6, 2/9, 9/13, 2/13, 2/12, 12/13, 10/13, 5/10, 2/5, 2/10, 10/11, 2/11, 2/4, 4/10, 1/10, 1/8, 2/8.